7/7/26

(1) Find out the list of companies which are conducting the placement drive using DSA course.

(2) Find out the syllabus of Gate for DSA

(1A) List of Companies - Google, Amazon, Microsoft, Adobe, Flipkart, Uber, Oracle, Goldman Sachs, Walmart Labs, Samsung R&D, TCS Digital, Infosys (Specialist roles), Directi, and Paytm.

(2A) GATE Syllabus for DSA

- Arrays, stacks, queues, linked lists
- Trees - binary trees, BST, balanced trees (AVL, B-trees)
- Graphs: representations, traversals (BFS/DFS)
- Hashing
- Sorting algorithms and their complexities
- Asymptotic notation (Big-O, Omega, Theta)

Algorithms portion (often paired with DS) also includes divide and conquer, greedy, dynamic programming, algorithm design and analysis.

Q1) Find out set of IIO for the set of following.

1. Electricity bill
2. Bus ticket
3. Message to a friend

Q2) Today when you started from home to clg what are the various datastructures you identified in your surroundings.

1 Ans

(i) Electricity Bill

Input: previous meter readings, current meter readings, number of units consumed, Tariff rate, Customer ID/acc number, taxes, fixed charges, billing period.

Output: Total units consumed, total amount payable, due date, bill number / reference, Itemized charges (energy charge + fixed charge + tax)

(ii) Bus Ticket

Input: Source and destination stop, date and time of travel, number of passengers, ticket type (student/general), fare rate per Km.

Output: Ticket number, total fare, seat number (if applicable), boarding point, printed/generated ticket.

(iii) Message to a friend

Input: Sender's message/text, recipient's contact number, Sender's network

Output: Message sent confirmation, Delivery status, Time of Sending

2 Ans

Data Structures around our Surroundings when we started from home to college

Array - Row of parked bikes/cars
Queue - Students waiting at the bus stop or canteen
Tree - Road map with branches
Graph - Road network connecting different places
Hash table - phone contacts stored by name
matrix - Seating arrangement in a classroom
Circular Queue - Round robin traffic at a signal.
Linked list - Chain of buses moving one after another
Stack - Stack of books or plates.